



# Cohort canonical evaluation — 20 agents

idea-01-lateral-attribution · 130 held-out segments · V0  
yaw 0.01456 rad/s, CTE 147.44 m · generated  
2026-05-31 23:15

- **Executive summary**

AGENTS (OK / TOTAL)

**20 / 20**

EVAL POOL

**130 segs**

V0 YAW RMSE

**0.01456**

V0 CTE RMSE

**147.44 m**

WALL TIME

**3.61s**

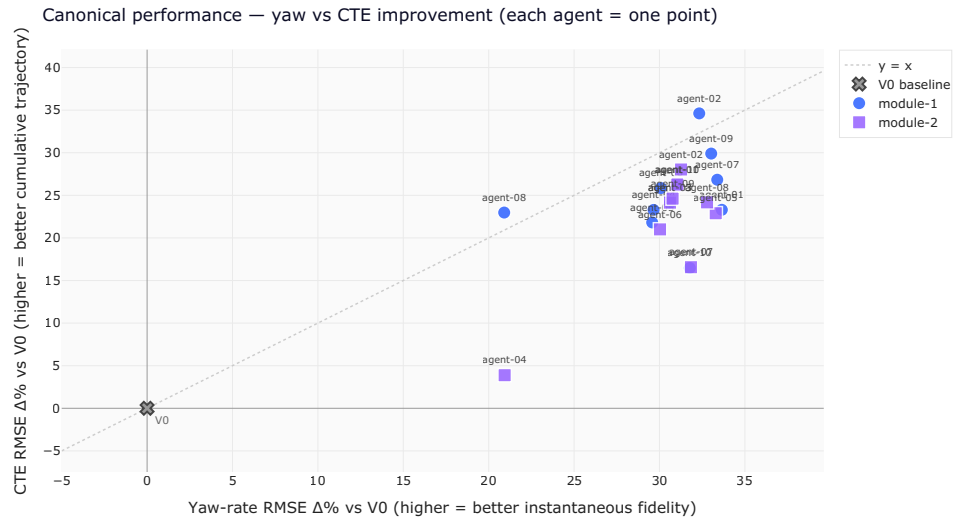
Best yaw: m1-agent-01 **+33.6%** · Best CTE: m1-agent-02

**+34.6%**

**Winning both KPIs  $\geq +30\%$  (1 agents): m1-agent-02**

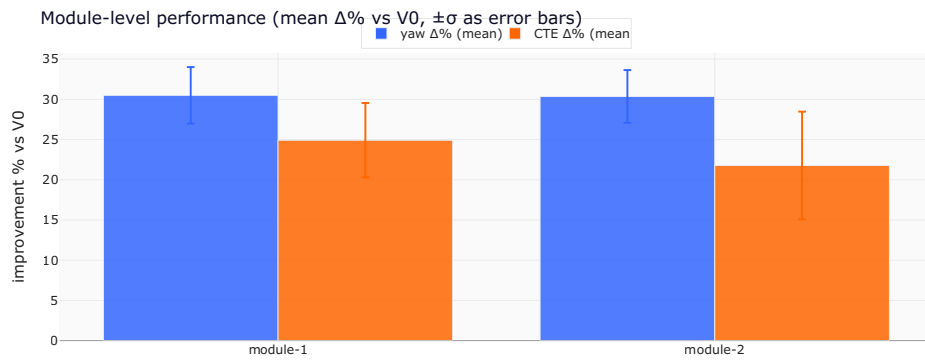
## 1. Headline scatter — yaw vs CTE improvement

Each agent is one point. **Top-right is the goal** — both KPIs improve over V0. The dashed diagonal is  $y = x$ : above it, an agent improved CTE more than yaw (good trajectory integration); below, the opposite. The V0 baseline sits at the origin.



## 2. Module-level aggregation

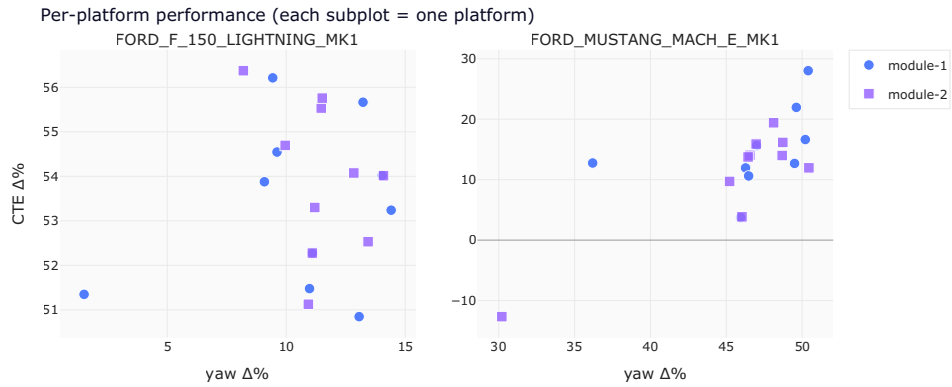
Performance grouped by which module shipped the agent.  
Mean  $\Delta\%$  over only the agents that reconstructed successfully;  
failures shown separately.



| family   | n ok / total | yaw $\Delta\%$ (mean $\pm \sigma$ ) | CTE $\Delta\%$ (mean $\pm \sigma$ ) |
|----------|--------------|-------------------------------------|-------------------------------------|
| module-1 | 10/10        | +30.5% $\pm$ 3.5%                   | +24.9% $\pm$ 4.5%                   |
| module-2 | 10/10        | +30.4% $\pm$ 3.3%                   | +21.8% $\pm$ 4.2%                   |

### • 3. Per-platform breakdown

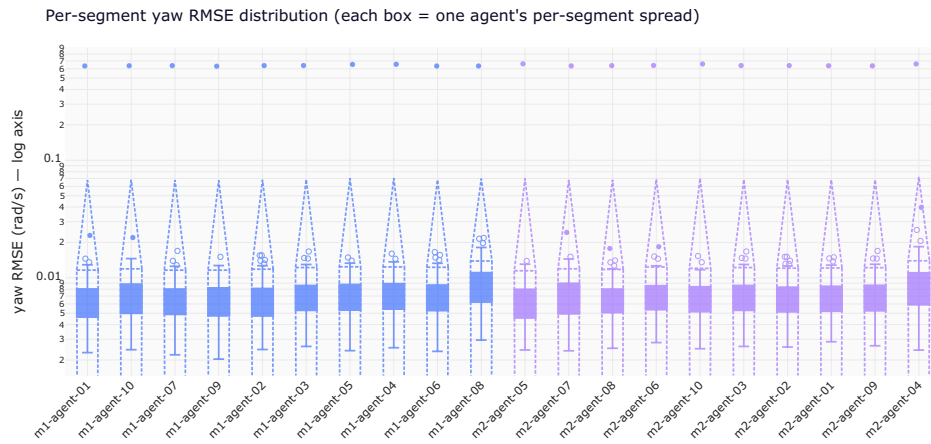
Same cohort, but split by the vehicle platform the agent was evaluated on. Agents that win one platform but lose another have a platform-specific calibration; agents whose dots cluster together generalise.



| platform                 | agents | yaw Δ% (mean ± σ) |
|--------------------------|--------|-------------------|
| FORD_F_150_LIGHTNING_MK1 | 20     | +11.1% ± 2.8%     |
| FORD_MUSTANG_MACH_E_MK1  | 20     | +46.3% ± 4.7%     |

## - 4. Per-segment yaw RMSE distribution

Each box summarises one agent's per-segment yaw RMSE — pooled RMSE can hide that an agent does great on most segments but pathologically badly on a few. The y-axis is logarithmic to compress outliers.



## 5. Per-agent canonical scorecard

The full table. Failed reconstructions are surfaced with their reason — no fake zeros.

| agent       | family   | status | yaw V0   | yaw final |
|-------------|----------|--------|----------|-----------|
| m1-agent-01 | module-1 | ok     | 0.014563 | 0.009662  |
| m1-agent-02 | module-1 | ok     | 0.014563 | 0.009855  |
| m1-agent-03 | module-1 | ok     | 0.014563 | 0.010105  |
|             | module-1 | ok     | 0.014563 | 0.010244  |

| agent       | family   | status | yaw V0   | yaw final |  |
|-------------|----------|--------|----------|-----------|--|
| m1-agent-04 |          |        |          |           |  |
| m1-agent-05 | module-1 | ok     | 0.014563 | 0.010256  |  |
| m1-agent-06 | module-1 | ok     | 0.014563 | 0.010181  |  |
| m1-agent-07 | module-1 | ok     | 0.014563 | 0.009701  |  |
| m1-agent-08 | module-1 | ok     | 0.014563 | 0.011518  |  |
| m1-agent-09 | module-1 | ok     | 0.014563 | 0.009753  |  |
| m1-agent-10 | module-1 | ok     | 0.014563 | 0.009936  |  |
| m2-agent-01 | module-2 | ok     | 0.014563 | 0.010049  |  |
| m2-agent-02 | module-2 | ok     | 0.014563 | 0.010010  |  |
| m2-agent-03 | module-2 | ok     | 0.014563 | 0.010104  |  |
| m2-agent-04 | module-2 | ok     | 0.014563 | 0.011514  |  |
| m2-agent-05 | module-2 | ok     | 0.014563 | 0.009714  |  |
|             | module-2 | ok     | 0.014563 | 0.010189  |  |

| agent       | family   | status | yaw V0   | yaw final |
|-------------|----------|--------|----------|-----------|
| m2-agent-06 |          |        |          |           |
| m2-agent-07 | module-2 | ok     | 0.014563 | 0.009925  |
| m2-agent-08 | module-2 | ok     | 0.014563 | 0.009788  |
| m2-agent-09 | module-2 | ok     | 0.014563 | 0.010082  |
| m2-agent-10 | module-2 | ok     | 0.014563 | 0.010041  |

## • 6. Calibration cards — fitted coefficients

Where the cohort converges on similar physics, and where it forks. Each tile is one agent's `coeffs.json` as shipped.

m1-agent-02

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.4760851016953556,
    "K_us": 0.0026722138032651867,
    "bias_rad": 0.0002686192499419401
```



```
},
"FORD_F_150_LIGHTNING_MK1": {
  "L": 3.809760457416018,
  "K_us": 0.0038412171248315657,
  "bias_rad": 0.001307379955568698
},
"HYUNDAI_IONIQ_5": {
  "L": 3.096229822470566,
  "K_us": 0.003695564361835472,
  "bias_rad": -0.0002960976826300522
},
"TESLA_MODEL_3": {
  "L": 2.875,
  "K_us": 0.0,
  "bias_rad": 0.0
}
}
```

---

## m1-agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": -0.004329037631466362,
    "b": 0.9177897626357783,
    "c": -0.00044178131403551616
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a": 0.0002206757945255914,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
  "HYUNDAI_IONIQ_5": {
    "a": 0.0020222989265629857,
    "b": 0.9120045284217976,
```

```
    "c": -0.00048449411444108955
  },
  "TESLA_MODEL_3": {
    "a": 0.0002206757945255914,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
  "_default": {
    "a": 0.0,
    "b": 1.0,
    "c": 0.0
  }
}
```

---

## m1-agent-06

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "best": "M2_understeer",
    "m1": {
      "a": 1.0870612234849137
    },
    "m2": {
      "L_eff": 2.4763030468069056,
      "K_us": 0.0024395167795073017
    }
  }
}
```

```
},
  "m3": {
    "a": 0.9751496691433119,
    "b": 0.23839497273290036,
    "c": 0.0004082870644824788
  },
  "val_rmse": {
    "V0": 0.0267999381028469,
    "M1": 0.027439581036525294,
    "M2": 0.026420879339395156,
    "M3": 0.031903168689021696
  },
  "train_rmse": {
    "V0": 0.012685628695945105,
    "M1": 0.011389013310887795,
    "M2": 0.008745921115146504,
    "M3": 0.010386863485176999
  }
},
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "best": "M2_understeer",
    "m1": {
      "a": 0.8569324797501493
    },
    "m2": {
      "L_eff": 3.882198880475702,
      "K_us": 0.003439279945738158
    },
    "m3": {
      "a": 0.8772240685582882,
      "b": -0.031353830652592465,
      "c": -0.0036354381262862774
    },
    "val_rmse": {
      "V0": 0.01591104278075153,
      "M1": 0.011141896721881754,
      "M2": 0.0073696198273352906,
      "M3": 0.0104650790814036
    },
    "train_rmse": {
      "V0": 0.02018452928972004,
```

## m1-agent-08

```
{
  "HYUNDAI_IONIQ_5": {
    "L": 3.0,
    "K": 0.0033895612010179177,
    "s": 0.9668278221152291,
    "off": 0.0005045046778856075,
    "lag": 3,
    "rmse": 0.00888482422851739
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "K": 0.002625595512351926,
    "s": 1.1771921499482927,
    "off": 3.9923690213217876e-05,
    "lag": 4,
    "rmse": 0.01315660341959157
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "K": 0.0035388455508032707,
    "s": 0.9569361755429138,
    "off": -0.0011908292677060655,
    "lag": 3,
    "rmse": 0.011832340064259452
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K": 0.0033895612010179177,
    "s": 1.0,
    "off": 0.0,
    "lag": 3,
    "rmse": null
  }
}
```

## m1-agent-10

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 2.5297190644305063,
    "K_us": 0.0015446536838391765,
    "d0": -4.756112262525493e-05,
    "tau_yaw": 0.02,
    "tau_steer": 0.05
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.834827559665228,
    "K_us": 0.0031524889056087045,
    "d0": 0.00115199881391885,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "HYUNDAI_IONIQ_5": {
    "L_eff": 3.1485752690608293,
    "K_us": 0.0021732037046845788,
    "d0": -0.0005619614330402095,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "TESLA_MODEL_3": {
    "_note": "no truth available \u2014 passthrough yaw_rate_pred.",
    "passthrough": true
  }
}
```

## m2-agent-03

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "platform": "FORD_F_150_LIGHTNING_MK1",
    "L": 3.7,
    "n_samples": 430990,
    "rmse_v0": 0.016327157090019005,
    "rmse_v1": 0.007867193883977044,
    "K_v1": 0.0043769531249999985,
    "rmse_v2": 0.00640773025816659,
    "K_v2": 0.003836687017912508,
    "a_v2": 0.9743406550510199,
    "b_v2": -0.001228592917784131
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "platform": "FORD_MUSTANG_MACH_E_MK1",
    "L": 2.984,
    "n_samples": 615175,
    "rmse_v0": 0.013616820108542612,
    "rmse_v1": 0.013930807470187448,
    "K_v1": 0.000785400390624989,
    "rmse_v2": 0.0095149036095479,
    "K_v2": 0.0028742800060782272,
```

```
"a_v2": 1.1998919734409932,  
"b_v2": 3.385609342398125e-05  
},  
"HYUNDAI_IONIQ_5": {  
  "platform": "HYUNDAI_IONIQ_5",  
  "L": 3.0,  
  "n_samples": 2042270,  
  "rmse_v0": 0.017084454875116246,  
  "rmse_v1": 0.009100616867655459,  
  "K_v1": 0.0042265624999999985,  
  "rmse_v2": 0.008741269113389604,  
  "K_v2": 0.0033859035164445724,  
  "a_v2": 0.9707125684779052,  
  "b_v2": 0.0005066770839987745  
}  
}
```

---

## m2-agent-07

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "tau": 0.05,  
    "L_eff": 3.8588781722695416,  
    "K_u": 0.00314039574481963,  
    "delta_bias_rad": -0.001157331673169034,  
    "rmse": 0.005592721659455879  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "tau": 0.08,  
    "L_eff": 2.5486304264900417,  
    "K_u": 0.0015312706121448161,  
    "delta_bias_rad": 4.8217638587095315e-05,  
    "rmse": 0.008347407485517062  
  },  
  "HYUNDAI_IONIQ_5": {  
    "tau": 0.05,
```

```
"L_eff": 3.163750474852756,  
"K_u": 0.0021607848879418314,  
"delta_bias_rad": 0.0005617097326308059,  
"rmse": 0.007713638558960502  
}  
}
```

- **6b. Operating-contract audit**

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The grader strips `sim_df` to an allowlist before calling each agent's `predict()`, so an agent cannot read truth at inference time. This section reports source-level intent — which agents *would have* read truth columns if the grader hadn't intervened. A clean audit is its own quality signal.

**Clean cohort:** no agent referenced truth-derived column names inside their `predict()` body.

## 7. Reconstruction quality (substrate signal)

How many agents shipped the right artefacts to be canonically gradable. Failures here are a substrate or contract problem, not a model problem.

| format check                       | pass | fail |
|------------------------------------|------|------|
| agent_folder_exists                | 20   | 0    |
| has_manifest_json                  | 20   | 0    |
| manifest_parsable                  | 20   | 0    |
| manifest_declares_predict_callable | 20   | 0    |
| manifest_declares_platform_support | 20   | 0    |

| format check    | pass | fail |
|-----------------|------|------|
| has_predict_py  | 20   | 0    |
| has_coeffs_json | 14   | 6    |
| has_report      | 3    | 17   |

## • 8. Worst-of-cohort

### Lowest yaw $\Delta\%$

- m1-agent-08  
(+20.9%)
- m2-agent-04  
(+20.9%)
- m1-agent-05  
(+29.6%)

### Lowest CTE $\Delta\%$

- m2-agent-04 (+3.9%)
- m1-agent-10  
(+16.5%)
- m2-agent-07  
(+16.6%)

Generated by grade-cohort-reports at 2026-05-31 23:15 · idea idea-01-  
lateral-attribution